ARIA equity research: validation summary

Point-in-time backtest of ARIA's forecasts, target prices and recommendations against what actually happened. Generated 27 Oct 2026, 00:37 (IST) from 4,779 NSE/BSE companies. Figures are as of this run and will move slightly as the last companies finish. Intervals are 95% unless stated, obtained by resampling whole companies (2,000 draws).

Overall verdict

Financial forecasting: modest improvement over naive. The base case beats 'carry the last value forward' by a skill of +0.10 at one year ahead (range -0.06 to +0.41 across revenue and EBITDA at 1, 3 and 5 years). Typical misses are large (12% of revenue at one year, 26% at five).

Recommendations: edge not demonstrated. The 12-month hit rate is 42.8% [41.8, 43.8], only +2.7pp (95% CI +1.7pp to +3.6pp) above random calls with the same mix, and 10.9pp below simply saying SELL on everything. BUY calls' median excess return is +2.4% [-0.3, +5.1], a CI that includes zero. On a common 3-class scoring ARIA is less accurate (36.0%) than always predicting the majority outcome (42.4%).

Ranking: weak signal. BUY > HOLD > SELL in median excess return at every horizon (at 3 months the differences are within the noise), and the BUY-minus-SELL spread is statistically positive (+22.9% [+19.3, +26.7] per 12 months, outliers clipped), but it is driven by a few years (FY2020-21), shrinks to roughly half in liquid stocks, and the rank correlation of upside with realised excess return is only 0.10.

1. Key findings

- **Financial forecasts (Experiment 1).** ARIA's base-case revenue forecast misses the later reported figure by a typical **12%** at one year ahead, **23%** at three and **26%** at five (median absolute log error). 'Skill vs naive' (section 3) says whether that is better than a trivial guess.
- **Scenario coverage.** The reported revenue fell between ARIA's bear and bull cases only 8-16% of the time. That is a description of the scenario spread, not a failed 68% calibration: the cases were never designed as a stated-probability interval (their spread equals ARIA's own Monte Carlo sigma, about +-1 sigma). Section 7 builds an empirically calibrated interval and tests it out of sample.
- **Recommendations (Experiment 2).** Across 2,460 companies the hit rates are 41.8% / 42.6% / 42.8% at 3 / 6 / 12 months, against 41.0% / 41.2% / 40.1% for random calls with the same mix. The edge over random is not significant at 3 months and small at 6 and 12.
- **Why 'always SELL' scores well.** 54% of stock-years trail the Nifty over 12 months: the median stock lags a cap-weighted index (a few big winners lift the mean, most stocks lag the median). So hit rates must be read against the base rate of the same event, not against 50%.
- **Target prices are far off.** Median absolute target-price error at 12 months is 42%, with targets sitting 22% below realised prices (ARIA's intrinsic value is usually below the market price).
- **Broker comparison.** No dataset of Indian brokerages' dated targets and ratings is available, so no head-to-head against brokers has been run (see section 8). Published academic studies are shown for context only.

2. Coverage

Outcome	Companies	Share
Scored (forecasts validated)	2,648	55.4%
Insufficient history (< 3 fiscal years / no valid origin)	1,161	24.3%
ARIA's valuation model declined (loss-making, no payout history, ...)	937	19.6%
Data source unreachable (to retry)	21	0.4%
No price history available	7	0.1%
No statements page	5	0.1%

Only 'Scored' companies enter the accuracy figures. Many NSE/BSE names are small, newly listed or delisted, so they lack the 3+ fiscal years of statements or price history a point-in-time test needs. Today's listings only, so survivorship flatters every result.

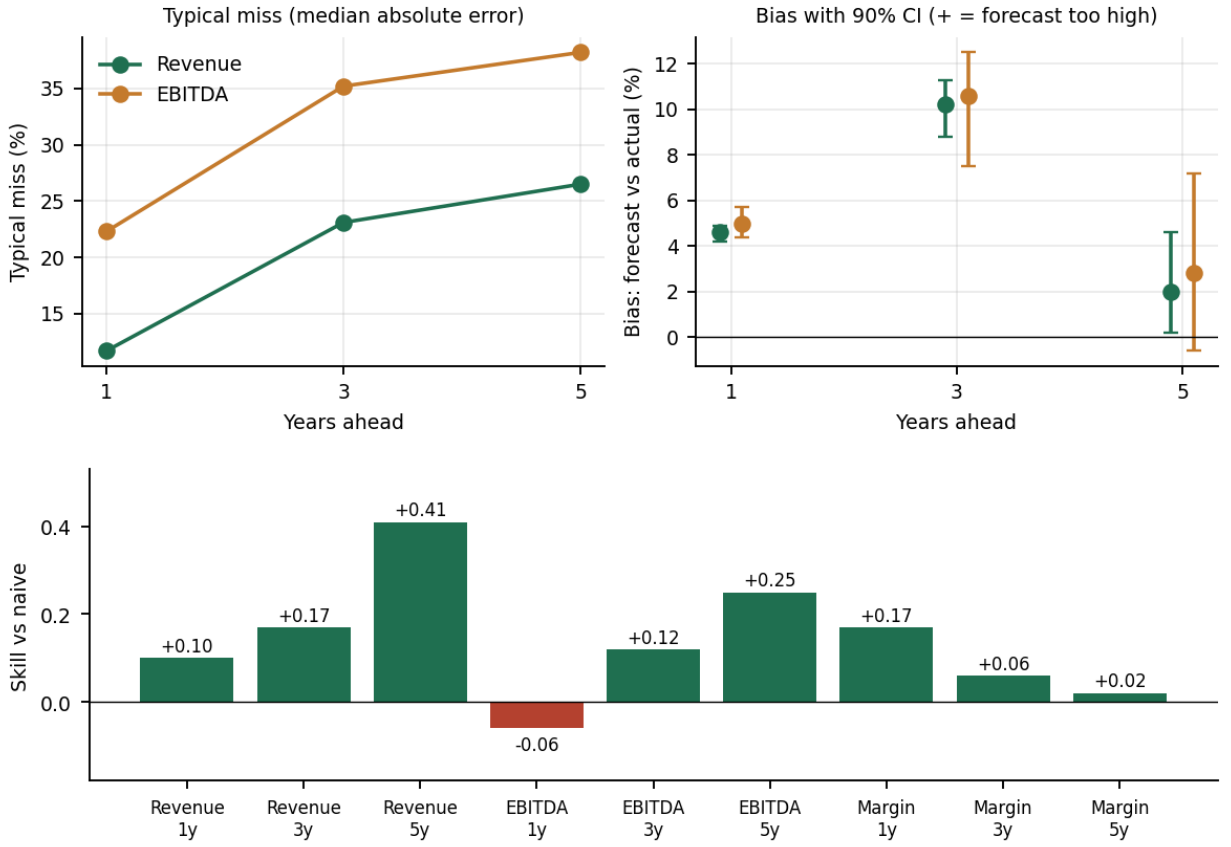
3. Experiment 1: how accurate are ARIA's financial forecasts?

For each past fiscal year-end (FY2018 to FY2025) ARIA's own bear/base/bull projections were rebuilt using only information public by then (annual results assumed available 90 days after year-end), then compared with what the company later reported. Typical miss = median absolute error of the base case.

How to read 'Skill vs naive'. A naive guess needs no model: *rw* carries the last reported value forward, *cagr* continues the recent growth rate, *mr* assumes the margin returns to its own average. Skill = 1 - (ARIA's typical miss / the naive guess's typical miss). **+0.20 means ARIA's typical error is 20% smaller than the naive one; 0 means no better; negative means worse.** At one year ahead revenue skill vs 'rw' is +0.10, i.e. ARIA is only about 10% better than repeating last year's figure; the advantage grows with the horizon (+0.41 at five years) because last year's value becomes a poor guess for a growing company.

Line	Years ahead	Companies	Typical miss	Bias (base)	Between bear and bull (scenario coverage)	Skill vs naive*
Revenue	1	1707	12%	+0.046	8%	rw +0.10
Revenue	3	1268	23%	+0.102	10%	rw +0.17
Revenue	5	758	26%	+0.020	16%	rw +0.41
EBITDA	1	1707	22%	+0.050	39%	rw -0.06
EBITDA	3	1268	35%	+0.106	29%	rw +0.12
EBITDA	5	758	38%	+0.028	29%	rw +0.25
EBITDA margin (pp)	1	1702	2.29	+0.31	43%	mr +0.17
EBITDA margin (pp)	3	1259	3.02	+0.66	34%	mr +0.06
EBITDA margin (pp)	5	750	3.27	+0.85	32%	mr +0.02
EPS (banks)	1	242	27%	+0.037	3%	rw +0.10
EPS (banks)	3	174	44%	-0.077	6%	rw +0.23
EPS (banks)	5	92	54%	-0.126	7%	rw +0.31

*Skill vs the main naive baseline (rw for levels, mr for margin). Bias is ln(forecast/actual): +0.10 means about 10% too high; margin bias is in percentage points. Ten-year horizons cannot be validated with 12 years of statements. 'Between bear and bull' is the share of reported figures that landed between the two cases.



Top: typical miss (left) and bias with 90% confidence intervals (right). Bottom: skill against the naive guess; green is better than naive, red worse.

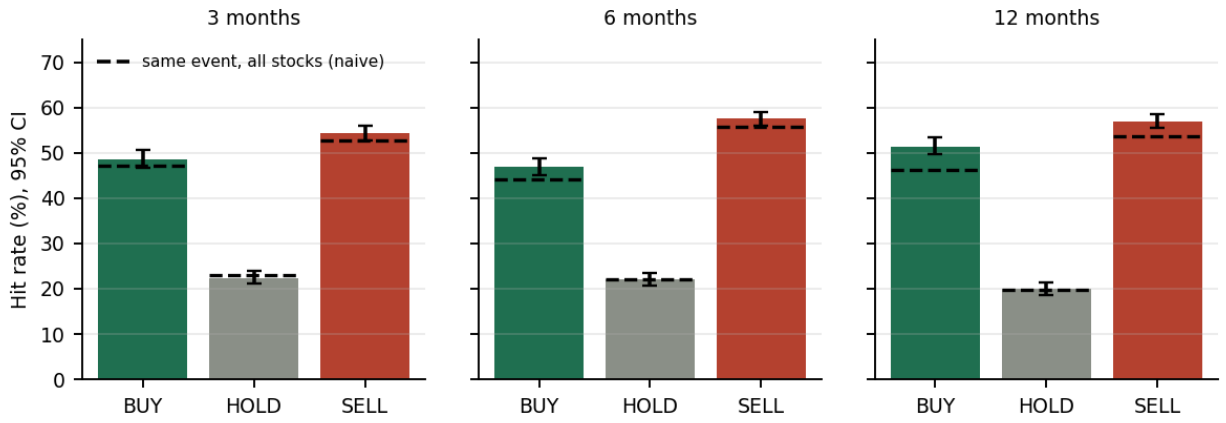
4. Experiment 2: would ARIA's report have worked as an investment call?

For every scored company and year, ARIA's own valuation rule (unchanged code) produced a target price and a BUY / HOLD / SELL call from point-in-time data. Outcomes use prices 3, 6 and 12 months later, total return (dividends included) versus the Nifty 50 price index. Rules were fixed before looking at results.

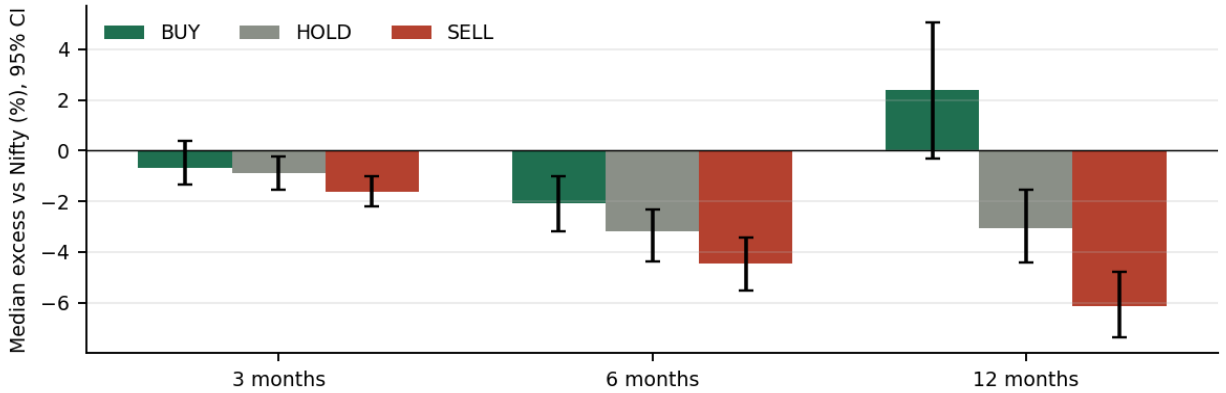
What counts as a success. BUY succeeds if the stock beats the Nifty (excess > 0). SELL succeeds if it trails the Nifty (excess < 0). **HOLD succeeds if the stock stays within +-5 / 7.5 / 10 percentage points of the Nifty** at 3 / 6 / 12 months, i.e. 'no material over- or under-performance'. BUY/SELL are sign tests and HOLD is a band test, so their hit rates are not comparable; each is shown against the base rate of the same event for all stocks. A single common scoring (3 classes, same threshold X) is given in section 5.

Mix of calls (12-month rows): SELL 41%, HOLD 35%, BUY 24%. Target price withheld by the model in about 58% of cases (the call is then made from the nearest edge of the fair-value range).

Measure	3 months	6 months	12 months
Rows (company-years)	9,981	9,981	9,981
ARIA hit rate [95% CI]	41.8% [40.8, 42.8]	42.6% [41.6, 43.6]	42.8% [41.8, 43.8]
Random calls, same mix	41.0%	41.2%	40.1%
Always BUY (= buy-and-hold)	47.2%	44.2%	46.3%
Always SELL	52.8%	55.8%	53.7%
Median excess return: BUY [95% CI]	-0.7% [-1.4, +0.4]	-2.1% [-3.2, -1.0]	+2.4% [-0.3, +5.1]
Median excess return: HOLD [95% CI]	-0.9% [-1.6, -0.2]	-3.2% [-4.4, -2.3]	-3.1% [-4.4, -1.5]
Median excess return: SELL [95% CI]	-1.6% [-2.2, -1.0]	-4.4% [-5.5, -3.5]	-6.1% [-7.4, -4.8]
Median |target - price| / price	37%	39%	42%
Rank corr. (upside vs excess return)	0.05	0.05	0.10



Hit rate of each call with 95% CI; the dashed line is the hit rate of the same event if every stock were scored (what a naive rule would get). A bar above its dashed line is the only sign of skill.



Median excess return after each type of call, with 95% CI (bars whose interval crosses zero are not distinguishable from the market).

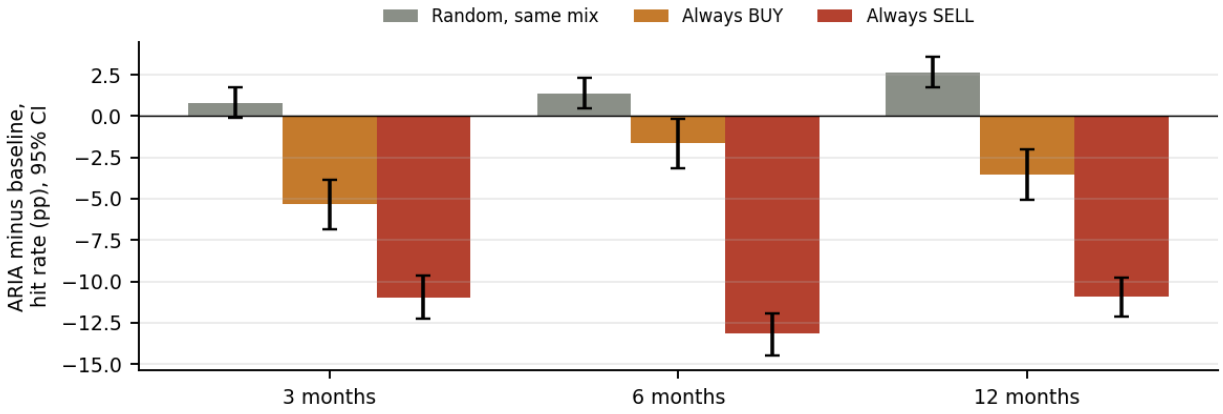
5. Statistical robustness

Method. Every ARIA-minus-baseline figure is a **paired** difference on the same rows. 95% CIs resample whole companies (2,000 draws, seed 0). p-values are clustered-bootstrap two-sided; a second test shuffles the calls among companies within each origin year (2,000 permutations). 'Random' draws calls 1,000 times with ARIA's own BUY/HOLD/SELL mix. Mean excess returns are dominated by a few small-cap multi-baggers, so mean-based figures below are also given with the excess return clipped at its 1st/99th percentile ('clipped'); medians are unaffected.

5.1 ARIA minus each baseline

ARIA minus ...	3 months	6 months	12 months
Hit rate (pp): random calls, same mix	+0.8 [-0.1, +1.7] p 0.093	+1.4 [+0.4, +2.3] p 0.006	+2.7 [+1.7, +3.6] p <.001
Hit rate (pp): always BUY (buy-and-hold)	-5.3 [-6.8, -3.8] p <.001	-1.6 [-3.1, -0.1] p 0.034	-3.6 [-5.1, -2.0] p <.001
Hit rate (pp): always SELL	-11.0 [-12.2, -9.7] p <.001	-13.2 [-14.5, -11.9] p <.001	-10.9 [-12.1, -9.8] p <.001
Hit rate (pp): always HOLD	+18.9 [+17.8, +20.0] p <.001	+20.5 [+19.4, +21.7] p <.001	+22.9 [+21.8, +24.0] p <.001
Signed excess return (pp, clipped): random calls, same mix	+1.4 [+1.0, +1.8] p <.001	+2.6 [+1.9, +3.3] p <.001	+6.9 [+5.6, +8.1] p <.001
Signed excess return (pp, clipped): buy-and-hold every stock	-2.9 [-3.6, -2.2] p <.001	-4.1 [-5.1, -3.0] p <.001	-9.5 [-11.1, -7.8] p <.001
Signed excess return (pp, clipped): always SELL	+4.4 [+3.8, +5.1] p <.001	+7.4 [+6.3, +8.5] p <.001	+18.6 [+16.8, +20.4] p <.001
Permutation p, hit rate vs same-year mix	0.396	0.001	0.001

Signed excess = +excess return for a BUY, -excess for a SELL, 0 for a HOLD (a long/short book of the calls). Always-SELL's positive signed return and higher hit rate reflect the base rate, see 5.3.



5.2 Each call separately (12 months)

Call	Rows (companies)	Hit rate [95% CI]	Same event, all stocks	Hit minus base (pp)	Median excess [95% CI]	Mean excess, clipped [95% CI]
BUY	2,429 (974)	51.4% [49.6, 53.4]	46.3%	+5.1 [+3.3, +6.8]	+2.4% [-0.3, +5.1]	+29.0% [+25.9, +32.4]
HOLD	3,480 (1,547)	20.0% [18.6, 21.5]	19.8%	+0.2 [-0.9, +1.3]	-3.1% [-4.4, -1.5]	+12.8% [+10.8, +14.8]
SELL	4,072 (1,685)	57.0% [55.5, 58.6]	53.7%	+3.3 [+2.2, +4.5]	-6.1% [-7.4, -4.8]	+6.1% [+4.3, +8.0]
BUY minus SELL						+22.9% [+19.3, +26.7]; raw +40.1% [+25.9, +64.2]

5.3 Why 'always SELL' scores 53-56%

Over 12 months **53.7%** of all stock-years trail the Nifty (95% CI 53.7% [52.7, 54.7]), 55.8% over 6 months and 52.8% over 3 months. The median stock returned 3.3% against 6.2% for the median Nifty window (median excess -3.3%), while the *mean* excess is 19.8%: returns are heavily right-skewed, a handful of multi-baggers lift the mean while most stocks lag a cap-weighted index. Share trailing the Nifty by size bucket: large 51%; mid 52%; small 55%; the universe is 71% small-cap, where lagging is most common. Adding dividends to the index (the TRI proxy) raises the share to 54.9%. Survivorship pushes the other way (failed stocks are missing), so the true base rate is if anything higher. Conclusion: a SELL hit rate of 53-56% is a base rate, not skill; the fair test is ARIA's SELL hit rate against the same event across all stocks.

5.4 Benchmark and return methodology

No free Nifty 50 Total Return Index series could be obtained: niftyindices.com's historical-data endpoint returned its HTML front page instead of data (request 1), and Yahoo Finance has no '^NSEITRI' symbol (404, request 2). A proxy is used instead: Nifty 50 price return plus a constant 1.3%/yr dividend yield. This is an assumption; the true TRI-price gap varied by year. The comparison below repeats the 12-month results under four return definitions; the qualitative conclusions are the same under each.

Excess-return definition (12 months)	Rows	ARIA hit rate [95% CI]	Hit minus random (pp)	Median excess: BUY / SELL	BUY minus SELL, mean [95% CI]
Total return vs Nifty price return (main run)	9,981	42.8% [41.8, 43.8]	+2.7 [+1.7, +3.6]	+2.4% [-0.3, +5.1] / -6.1% [-7.4, -4.8]	+40.1% [+25.9, +64.2]
Price return vs price return (like for like)	9,981	42.6% [41.6, 43.6]	+2.4 [+1.5, +3.4]	+0.7% [-1.0, +3.4] / -6.7% [-8.0, -5.6]	+39.4% [+25.3, +63.5]
Total return vs Nifty TRI proxy (+1.3%/yr)	9,981	42.8% [41.9, 43.8]	+2.8 [+1.9, +3.7]	+0.8% [-1.6, +3.5] / -7.4% [-8.7, -6.1]	+40.0% [+25.9, +64.1]
Beta-adjusted alpha vs TRI proxy	9,149	43.2% [42.2, 44.3]	+3.2 [+2.2, +4.1]	+3.3% [+0.4, +5.7] / -6.6% [-8.2, -5.6]	+43.0% [+27.5, +67.3]
Main run, clipped at 1st/99th percentile	9,981	42.8% [41.8, 43.8]	+2.7 [+1.7, +3.6]	+2.4% [-0.3, +5.1] / -6.1% [-7.4, -4.8]	+22.9% [+19.3, +26.7]

Alpha = stock total return - [rf + beta x (TRI proxy - rf)] with a constant 6.5% risk-free rate and a raw OLS beta on the 260 weekly returns before each origin (cached prices; available for 92% of rows). The price-return and TRI-proxy versions bracket the headline numbers.

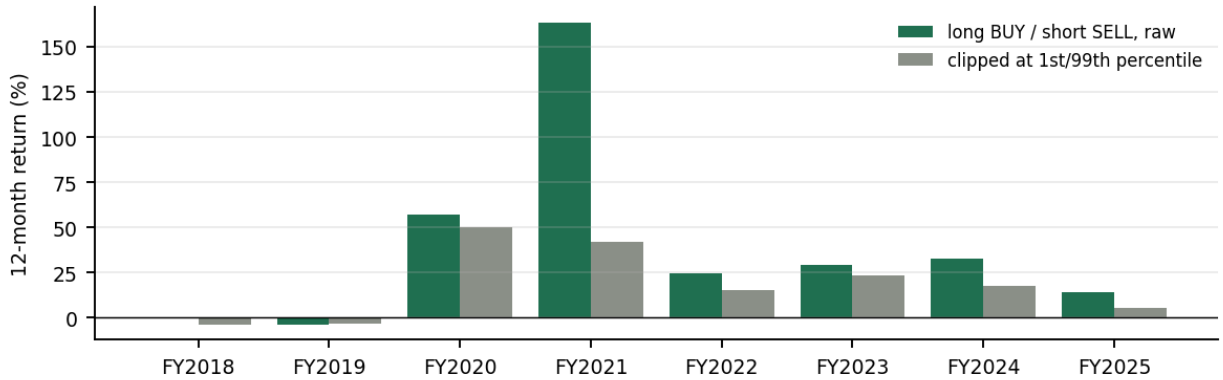
5.5 One common outcome definition: 3-class accuracy (12 months)

ARIA call \ realised	Outperform (> +10pp)	In line	Underperform (< -10pp)	Precision
BUY	1,070	420	939	44.1%
HOLD	1,318	697	1,465	20.0%
SELL	1,381	862	1,829	44.9%
Recall	28.4%	35.2%	43.2%	

Realised class uses the same X as the HOLD band, fixed in advance. Class shares: outperform 38%, in-line 20%, underperform 42%. **Accuracy 36.0% [35.1, 37.0]; always predicting the majority class ('underperform') gets 42.4%; random calls with ARIA's mix 33.4%. ARIA minus majority baseline: -6.4 [-7.6, -5.2] p <.001.** At 3 and 6 months the same gap is -6.2 [-7.5, -5.0] and -8.5 [-9.8, -7.2]. ARIA beats random assignment but not the trivial majority-class rule, because the market's skew makes 'underperform' the most common outcome.

5.6 Portfolio-style evaluation (12 months, equal weight, rebalanced yearly)

Long every BUY and short every SELL at each origin year (hypothetical: borrowing is unavailable for most Indian cash stocks); also long-BUY-only against the Nifty. Costs are charged as a round-trip rate on the fraction of the book replaced (turnover; the first year is a full build). The cost-adjusted CI is the gross bootstrap CI shifted by the average cost.



Strategy / period	Years > 0	Avg turnover	Gross, raw	Net of 0 bps (clipped)	25 bps	50 bps	100 bps
Long-short, all years	6/8	118%	39.7%	18.3% [14.5, 22.2]	18.0% [14.2, 21.9]	17.7% [13.9, 21.6]	17.1% [13.4, 21.0]
Long-short, FY2018-2021	2/4	131%	54.2%	21.2% [15.6, 26.8]	20.9% [15.3, 26.5]	20.6% [15.0, 26.2]	19.9% [14.3, 25.5]
Long-short, FY2022-2025	4/4	106%	25.1%	15.4% [9.9, 21.1]	15.1% [9.6, 20.9]	14.9% [9.4, 20.6]	14.3% [8.9, 20.1]

Strategy / period	Years > 0	Avg turnover	Gross, raw	Net of 0 bps (clipped)	25 bps	50 bps	100 bps
Long BUY vs Nifty, all years	7/8	61%	49.4%	26.1% [23.0, 29.4]	26.0% [22.9, 29.2]	25.8% [22.7, 29.1]	25.5% [22.4, 28.8]
Long BUY vs Nifty, FY2018-2021	3/4	65%	59.4%	25.6% [21.4, 30.2]	25.4% [21.2, 30.1]	25.2% [21.1, 29.9]	24.9% [20.7, 29.6]
Long BUY vs Nifty, FY2022-2025	4/4	56%	39.4%	26.7% [22.0, 31.9]	26.6% [21.9, 31.8]	26.4% [21.7, 31.6]	26.2% [21.5, 31.3]

Costs barely move the result because the gross spread is large relative to 25-100 bps x turnover; small-cap round trips can cost more than 100 bps and are not modelled beyond this sweep. Yearly figures swing from -6% to +171% (raw): the average is driven by FY2020-21.

Liquidity filter and chronological split

12-month universe	Rows	Hit rate [95% CI]	Hit minus random (pp)	BUY minus SELL, raw [95% CI]	Long-short, clipped, 50 bps
all stocks	9,981	42.8% [41.8, 43.8]	+2.7 [+1.7, +3.6]	+40.1% [+25.9, +64.2]	+17.7% [+13.9, +21.6]
traded value >= INR 1 cr/day	4,736	43.8% [42.3, 45.2]	+2.3 [+1.1, +3.6]	+22.0% [+16.0, +28.8]	+11.2% [+6.0, +16.3]
market cap >= INR 1,000 cr	4,840	43.6% [42.2, 44.9]	+2.1 [+0.9, +3.3]	+22.0% [+15.5, +29.2]	+10.5% [+5.5, +15.6]

Chronological split	Rows	Hit rate [95% CI]	Hit minus random (pp) [CI]	BUY minus SELL [95% CI]	3-class accuracy minus majority (pp)
3m, FY2018-21	3,818	40.3% [38.7, 41.8]	+0.5 [-1.0, +1.9]	+5.9% [+2.9, +9.2]	-15.8 [-18.0, -13.4]
3m, FY2022-25	6,163	42.8% [41.5, 44.2]	+2.3 [+1.1, +3.6]	+11.4% [+7.8, +15.7]	-6.9 [-8.8, -4.9]
6m, FY2018-21	3,818	39.3% [37.8, 40.9]	-0.5 [-2.0, +1.1]	+9.1% [+3.5, +15.3]	-16.9 [-19.2, -14.7]
6m, FY2022-25	6,163	44.7% [43.4, 45.9]	+3.4 [+2.2, +4.5]	+20.1% [+14.8, +26.2]	-3.3 [-4.8, -1.7]
12m, FY2018-21	3,818	41.7% [40.2, 43.2]	+2.8 [+1.3, +4.2]	+48.6% [+23.1, +91.7]	-8.3 [-10.4, -6.2]
12m, FY2022-25	6,163	43.4% [42.1, 44.7]	+2.8 [+1.6, +4.0]	+33.4% [+22.9, +45.1]	-5.2 [-6.7, -3.7]

Nothing is fitted on either half (rules and bands were fixed in advance), so this tests stability over time. The edge over random is absent in FY2018-21 at 3 and 6 months and appears in FY2022-25; the BUY-minus-SELL spread is positive in both halves. The 68%-interval scale in section 7 is the only quantity fitted on early data.

6. Data controls

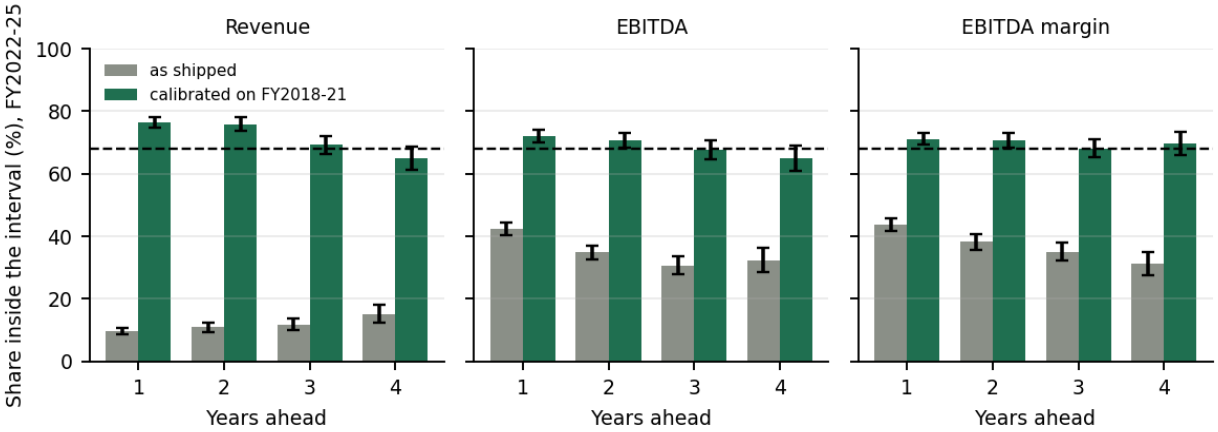
Not controlled: delisted and failed companies are absent (survivorship flatters every strategy); statements are screener.in's restated figures with an assumed 90-day publication lag; sector labels and the risk-free rate, equity risk premium and terminal growth are today's values applied to the past; the Nifty TRI is a proxy; overlapping origins share market shocks that company-level bootstrapping cannot capture; no borrow, impact or tax is modelled.
Quantified: 35% of origins have no usable sector label; 26% are flagged structural break and 10% short history; the size mix is 71% small, 22% mid, 7% large.

12-month slice	Rows	Hit rate [95% CI]	Hit minus random (pp)	BUY minus SELL, raw [95% CI]
all rows	9,981	42.8% [41.8, 43.8]	+2.7 [+1.7, +3.6]	+40.1% [+25.9, +64.2]
excluding structural-break and short-history origins	6,616	42.5% [41.3, 43.7]	+2.7 [+1.6, +3.8]	+40.2% [+20.9, +72.8]
classified sector only	6,447	43.5% [42.2, 44.7]	+2.9 [+1.8, +4.0]	+28.8% [+22.7, +35.1]
large-cap only	707	46.7% [43.1, 50.4]	+1.1 [-1.8, +4.0]	+9.8% [+1.7, +18.6]
mid-cap only	2,148	43.4% [41.4, 45.4]	+2.3 [+0.6, +4.0]	+17.8% [+8.2, +27.1]
small-cap only	7,107	42.2% [41.1, 43.3]	+3.4 [+2.3, +4.5]	+42.5% [+25.4, +68.1]

Dropping structural-break and short-history origins leaves both the hit-rate edge and the spread unchanged; dropping unclassified sectors leaves the hit-rate edge but shrinks the raw spread. The signal is strongest in small caps and weak in large caps (hit minus random not significant).

7. Scenario coverage and an empirically calibrated interval

The bear and bull cases were not designed as a stated-probability interval; their spread equals ARIA's own Monte Carlo sigma, so they behave like a roughly +-1 sigma scenario range. 'Scenario coverage' (the share of reported figures between the cases) is therefore a description, not a calibration failure of a 68% interval. To give a usable interval, the log-width of [bear, bull] around the base case is multiplied by a factor chosen on early origins (FY2018-21) so that 68% of reported figures fall inside, and that interval is then scored on FY2022-25 origins it never saw.



Line	Years	Scale s (fit FY18-21)	Test rows	As shipped [95% CI]	Calibrated [95% CI]	Single s per line [95% CI]
Revenue	1	13.5x	3,323	10% [9, 11]	77% [75, 78]	63% [61, 65]
Revenue	2	11.0x	2,220	11% [9, 12]	76% [74, 78]	70% [67, 72]
Revenue	3	8.5x	1,328	12% [10, 14]	69% [66, 72]	74% [71, 76]
Revenue	4	6.7x	598	15% [12, 18]	65% [61, 69]	79% [76, 83]
EBITDA	1	2.4x	3,163	42% [40, 44]	72% [70, 74]	75% [73, 77]
EBITDA	2	2.8x	2,093	35% [33, 37]	71% [68, 73]	69% [67, 71]
EBITDA	3	2.9x	1,240	31% [28, 34]	68% [65, 71]	66% [63, 69]
EBITDA	4	2.7x	553	32% [28, 36]	65% [61, 69]	64% [60, 68]
Margin	1	2.3x	3,326	44% [42, 46]	71% [69, 73]	75% [73, 77]
Margin	2	2.7x	2,224	38% [36, 41]	71% [68, 73]	70% [68, 73]
Margin	3	2.7x	1,331	35% [32, 38]	68% [65, 71]	67% [64, 70]
Margin	4	2.8x	599	31% [28, 35]	70% [66, 73]	67% [63, 70]
EPS (banks)	1	27.0x	436	5% [3, 7]	70% [64, 75]	58% [52, 63]
EPS (banks)	2	18.6x	254	9% [6, 14]	69% [62, 75]	65% [58, 72]
EPS (banks)	3	14.0x	132	11% [5, 18]	70% [62, 79]	73% [64, 81]

Scale s multiplies the log distance from base to each case (margin: the percentage-point distance); s = 1 is the shipped cases. Revenue cases are very tight (s of 6-13x needed) because they only vary the growth path slightly, whereas EBITDA and margin cases need about 2-3x. Out of sample the calibrated interval covers close to the 68% target (typically within about 9 points), where the shipped cases cover 5-45%. One scale per line over all horizons (last column) is worse than a scale per horizon. 16,568 observations were dropped because the cases were not ordered bear < base < bull. Years 5 and EPS beyond 3 years have too few later origins to score.

8. Comparison with brokerages

What can and cannot be claimed. A fair head-to-head needs brokerages' own dated targets and ratings for the same stocks and dates. No free, permitted dataset of that kind was found, and nothing was scraped or invented. The table below puts ARIA's results next to **published academic studies of analyst target-price accuracy**. These use different markets, periods, sample selections and definitions, so they are context, not a ranking.

Measure	ARIA (this study, 12 months)	Published evidence (analysts)
Share of targets met at 12 months	33.6% of upside targets (price at or above target); ARIA's targets are often below the market price	38% (US, 2000-2009, Bradshaw, Brown & Huang 2013); 63% of targets achieved within 12 months (Indian sample, 2024 study, per its abstract)
Average absolute target-price error	median 42%, mean 51%	about 45% on average (US, Bradshaw, Brown & Huang 2013)
Implied return minus actual return	mean -30 pp, median -22 pp (forecast minus actual; negative = ARIA's implied return was below what happened)	analysts' implied returns exceeded actual by about 15 pp on average (US)
Recommendation hit ratio	BUY 51% [50, 53], overall 43% [42, 44]	hit ratio not above 50% in a study of 1,000+ Indian recommendations (IIM Bangalore); analysts skew to BUY

Sources: Bradshaw, Brown & Huang (2013), *Do sell-side analysts exhibit differential target price forecasting ability?*, Review of Accounting Studies 18(4), ideas.repec.org/a/spr/reaccs/v18y2013i4d10.1007_s11142-012-9216-5.html; Indian target-price accuracy study (Cogent Economics & Finance, 2024), ideas.repec.org/a/taf/oaefxx/v12y2024i1p2423261.html; IIM Bangalore, *Investment value of analyst recommendations: evidence from the Indian stock market*, research.iimb.ac.in/fac_pubs/406. Figures are taken from abstracts/summaries and should be checked against the full papers before citing.

To run a real head-to-head, supply a CSV of brokerage calls (columns: symbol, date, broker, rating, target, horizon) and run `python scripts/scenario_report_validation.py --brokers broker.csv`. It scores the brokers with the same outcome rules and prints them beside ARIA on the same symbols.

9. Limitations

- Survivorship: only today's listed companies are included; delisted failures are missing, which flatters both ARIA and any benchmark.
- Statements are restated figures from screener.in (original as-first-reported values are not available) with an assumed 90-day publication lag; sector labels and risk-free/equity-premium inputs are today's.
- Experiment 2 tests ARIA's intrinsic-value call only; the live report also uses peer multiples and an LLM debate that cannot be reconstructed historically.
- Overlapping origins and horizons make the effective sample much smaller than the row counts; confidence intervals resample companies, not calendar time, so they are if anything too narrow (see the yearly spread in 5.6).
- The Nifty TRI could not be downloaded; a price-plus-1.3%-dividend proxy is used and results are shown under four return definitions.
- Published-study figures come from different markets and definitions and are shown as context, not as a like-for-like comparison.